

THEOPHILUS THOBEJANE

AI Engineer | LLM & Agentic Systems Specialist

Kempton Park, Gauteng | thobejanetheo@gmail.com | +27 66 360 3021 | **LinkedIn:** [linkedin.com/in/theophilusthobejane](https://www.linkedin.com/in/theophilusthobejane)
| **GitHub:** github.com/TheoInCodeLand | **Portfolio:** theophilus-portfolio.vercel.app | **Availability:** Immediate

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building production-grade LLM systems and agentic solutions using LangChain, FastAPI, and Node.js. Proven track record developing RAG architectures, conversational AI agents, and document verification workflows for enterprise clients. Recent ICT graduate (Advanced Diploma) passionate about transforming banking, insurance, and healthcare workflows through intelligent automation. Based in Johannesburg area and available for immediate start.

EDUCATION

Advanced Diploma in Information and Communication Technology (Applications Development)
University of Mpumalanga, Mbombela | **Completed: November 2025**

Diploma in Information and Communication Technology (Applications Development)
University of Mpumalanga, Mbombela | **Completed: December 2023** | **Formally Awarded: 3 May 2024**

Certifications: IBM Applied Data Science with Python, IBM Agile Explorer (Scrum/Kanban), IBM Data Fundamentals

TECHNICAL SKILLS

AI & Large Language Models: LLM integration (GPT-4, Claude, Groq, Gemini), RAG architecture, LangChain patterns, agentic solutions, semantic search, embedding models (sentence-transformers), prompt engineering, fine-tuning workflows

Backend Development: Python (FastAPI), Node.js (Express), REST API design, microservices architecture, webhook handling, ETL pipelines

AI Frameworks & Tools: LangChain, AutoGen patterns, Crew.ai concepts, FastEmbed, Pinecone Vector DB, Groq LLM, OpenAI API, Google Generative AI

Database & Cloud: PostgreSQL, SQLite, Pinecone, Cloudinary CDN, Vercel/Serverless deployment, Firestore

Document Processing: PDF extraction (pdfplumber, pdf-parse), OCR workflows, document verification pipelines, risk assessment algorithms, mammoth (DOCX processing)

DevOps & Security: Playwright web scraping, HMAC authentication, session management, QR generation, API security

Professional Competencies: Strategic thinking, Innovation-Driven, Cross-Functional Collaboration

PROJECTS & ENTERPRISE SOLUTIONS

AI Engineer | Full-Stack AI Developer

Freelance & Personal Projects | 2024 – Present

Axiora AI – Enterprise Knowledge Base & Conversational AI Platform

- Architected multi-tenant RAG system using FastAPI, Pinecone, and Groq LLM with 4-stage retrieval pipeline achieving 94% relevance accuracy on 10,000+ documents
- Developed "Casi" AI Agent with emotional intelligence layer and topology classification (COMPARATIVE_MATRIX, SEQUENTIAL_PROCEDURE, BROWSABLE_INVENTORY) for context-aware enterprise responses
- Implemented semantic search engine with two-stage retrieval (high-recall fetch + cross-encoder reranking) and HMAC-based service authentication
- Insurance/Banking Application: Conversation state management and document ingestion pipeline directly transferable to underwriting agents and customer due diligence systems

JobTrack – AI-Powered Application Tracking & ATS Optimization System

- Built dual-stack architecture (Node.js/Express + Python/FastAPI) with LLM-powered ATS analyzer using dual-mode PDF extraction
- Developed ML scoring algorithm correlating resume tailoring (0.42), referrals (0.38), and experience fit (0.31) with interview success
- Automated CV skill extraction and job requirement matching using 200+ technology taxonomy, reducing manual analysis time by 75%
- Insurance Parallel: Document parsing and risk-scoring methodology applicable to claims document analysis and automated underwriting logic

Casalinga Tours – AI Concierge for Travel Booking

- Integrated Google Gemini 2.5 Flash with real-time inventory injection for contextual tour recommendations using Node.js backend
- Automated 70% of pre-booking inquiries through intelligent routing and booking intent detection, reducing response time from hours to seconds
- Implemented multi-modal processing with image recognition for tour galleries and automated content tagging
- Insurance/Healthcare Parallel: Customer inquiry classification and automated response generation applicable to claims triage and pharmacy workflow optimization

Before You Sign – Vehicle Verification & Risk Assessment Platform

- Developed 7-step document verification workflow system with audit trails (certification, inspection, dispute resolution) using Node.js and PostgreSQL
- Implemented risk scoring algorithm with weighted confidence scoring for vehicle authenticity and QR-based verification with cryptographically signed codes
- Insurance Application: Verification workflow directly applicable to automated underwriting, claims verification, and fraud pattern detection

REFERENCES

- **Ms. Sinazo Matyila** (Lecturer, University of Mpumalanga) | sinazo.matyila@ump.ac.za | (074) 336 5336
- **Blessing Abrahams** (Promoter, Casalinga Tours) | +27 76 687 8485
- **Mxolisi Mandlenkosi Mkhonza** (Founder, Before You Sign) | mxolisi@beforeyousign.co.za | +27 83 744 0206